A product counterexample to the simplex maximum for projection-body volume

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Abstract

The product of two ten-dimensional simplices has larger normalized projection-body volume than a twenty-dimensional simplex. This gives a counterexample to Brannen's proposed simplex maximum.

1 Introduction

Throughout, a convex body in $\mathbb{R}^d$ is compact and convex with nonempty interior, and $d \geq 2$. Write $|K|$ for its d-dimensional volume and $h_C(x) = \max_{y \in C} \langle x, y \rangle$ for the support function of a compact convex set C. The projection body $\Pi_d K$, also written ΠK when the dimension is evident, is defined by

$$h_{\Pi K}(u) = \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} K), \qquad u \in S^{d-1},$$

extended homogeneously to all of $\mathbb{R}^d$. Here S^{d-1} is the Euclidean unit sphere and $\mathrm{proj}_{u^\perp}$ is orthogonal projection. We write B_2^j for the Euclidean unit ball in $\mathbb{R}^j$ and $\kappa_j = |B_2^j|$. The functional under consideration is

$$R_d(K) = \frac{|\Pi K|}{|K|^{d-1}}, \qquad c_d = \frac{(d+1)d^d}{d!}. \tag{1}$$

The extremal values of this affine-invariant functional are classical questions in convex geometry. Petty established the affine covariance of the projection-body operator [6]; see also [5, Equation (1)]. Brannen conjectured in 1996 that simplices maximize R_d among all convex bodies [1]. The simplex value is c_d, as we verify below. Thus the proposed simplex upper bound is

$$|\Pi K| \leq c_d |K|^{d-1}. \tag{2}$$

It is the volume of ΠK itself that occurs here, not the volume of its polar: the classical projection inequalities involving the polar concern a different functional. For $d \geq 3$, Petty's separate minimum problem asks whether ellipsoids are the only minimizers of the same functional R_d; see [7, Section 1] for these distinctions.

In dimension three, Chen, Feng, Li, Xi, and Xu settle the minimum problem and also prove the bound $R_3(K) \leq 18 = c_3$ for all three-dimensional convex bodies, with tetrahedra attaining equality [2, Theorems 1.1 and 1.2]. Feng, Hu, Liu, and Xu give counterexamples to (2) in every dimension $d \geq 9$, using polytopes with at most $d+2$ facets [3, Theorem 1.3]. Their result already gives a negative answer to the unrestricted-dimensional simplex-maximum question. Their construction uses perturbed projections of a cube and Minkowski's existence theorem [3, Section 4]. The present paper supplies a direct Cartesian-product construction and its exact value: two ten-dimensional simplices suffice.

Theorem 1. *Let $T_{10} = \operatorname{conv}(0, e_1, \dots, e_{10}) \subset \mathbb{R}^{10}$, where the e_i are the coordinate unit vectors, and let $K = T_{10} \times T_{10} \subset \mathbb{R}^{20}$. Then*

$$\frac{R_{20}(K)}{c_{20}} = \frac{121\binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096} > 1.$$

Thus K violates (2) in dimension twenty.

The exact witness in Theorem 1 provides an elementary illustration of the failure of the simplex maximum. The optimal upper constant and its maximizing bodies remain separate questions.

The useful mechanism is a product test: for full-dimensional polytopes $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$, with $r, s \geq 2$,

$$R_{r+s}(A \times B) = R_r(A) R_s(B).$$

Consequently, the product $c_r c_s$ of the simplex values is a necessary lower bound for any universal upper constant in dimension $r + s$. The dimension-twenty counterexample comes from comparing two copies of the ten-dimensional simplex value with the twenty-dimensional simplex value. All identities used in that comparison are proved below.

The growth of upper constants with dimension is a further quantitative question. For centrally symmetric bodies, Henk gives the lower bound $2^d(9/8)^{\lfloor d/3 \rfloor}$ for the optimal upper constant [4]. Iterating our product identity gives an exponential excess over the simplex benchmark: for some fixed $\lambda > 1$ and every sufficiently large n, a product of simplices $K_n \subset \mathbb{R}^n$ satisfies $R_n(K_n) \geq \lambda^n c_n$. Corollary 6 proves this consequence by repeating ten-dimensional factors and using one additional simplex to reach each dimension.

Section 2 derives the facet formula, the product identity, and affine covariance. Section 3 computes the simplex value by expressing its projection body as a cube plus one segment. Section 4 combines the identities and gives the exact integer comparison, then proves the exponential excess in all sufficiently large dimensions.

2 Facet vectors and Cartesian products

We first express projection bodies of polytopes as sums of segments. The facets of a Cartesian product then separate into two families, which will make its normalized projection-body volume multiplicative. The facet formula is the polytopal case of Cauchy's projection formula; see [7, Section 1]. We include its proof to fix the factor $1/2$ on which the subsequent constants depend.

We use Minkowski addition $C + D = \{c + d : c \in C,\ d \in D\}$. Support functions add under this operation and determine compact convex sets uniquely. For the latter assertion, if $x \notin C$ and $y \in C$ is nearest to x, minimization along each segment $[y, z] \subset C$ gives $\langle x - y, z - y \rangle \leq 0$ for every $z \in C$. The linear functional with vector $x - y$ therefore separates x from C.

Lemma 2. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a full-dimensional convex polytope. For each facet F, let s_F be its $(d-1)$-dimensional volume and ν_F its outward unit normal. Then*

$$h_{\Pi P}(u) = \frac{1}{2} \sum_F s_F |\langle \nu_F, u \rangle|, \qquad u \in \mathbb{R}^d, \tag{3}$$

and therefore

$$\Pi P = \sum_F \left[-\frac{s_F \nu_F}{2}, \frac{s_F \nu_F}{2} \right]. \tag{4}$$

Proof. First take u to be a unit vector. Orthogonal projection from the hyperplane of F to $u^\perp$ has absolute Jacobian $|\langle \nu_F, u\rangle|$. Indeed, for an orthonormal tangent basis $v_1, \dots, v_{d-1}$, that Jacobian is $|\det(v_1, \dots, v_{d-1}, u)|$, which equals the stated inner product in absolute value.

Apart from a set of $(d-1)$-dimensional measure zero, a point in the projection of P lies on the projections of exactly two facet interiors: the entry and exit points of its line parallel to u. To see that the exceptions are null, discard the boundary of the projection and the projections of all faces of dimension at most $d-2$. Also discard projections of facets whose hyperplanes are parallel to u; these projections have dimension at most $d-2$. Every remaining line meets P in a nondegenerate interval with its two endpoints in facet interiors. Integrating this multiplicity gives (3) for unit u, and homogeneity gives the general case.

The support function of $[-v/2, v/2]$ is $\frac{1}{2}|\langle v, u\rangle|$. Adding these support functions proves (4). □

Proposition 3. *Let $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$ be full-dimensional convex polytopes, where $r, s \geq 2$. In the orthogonal product space $\mathbb{R}^{r+s} = \mathbb{R}^r \times \mathbb{R}^s$,*

$$\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B). \tag{5}$$

In particular,

$$R_{r+s}(A \times B) = R_r(A) R_s(B). \tag{6}$$

Proof. The face exposed by a linear functional (u, v) on $A \times B$ is the product of the faces exposed by u and v. Dimensions add under Cartesian products. A nonzero linear functional exposes a proper face of a full-dimensional body, so if both u and v are nonzero, the product face has codimension at least two. A facet therefore has one facet factor and one whole factor. Thus the facets are $F \times B$ and $A \times G$, with area-normal vectors

$$(|B| s_F \nu_F, 0) \quad \text{and} \quad (0, |A| s_G \nu_G),$$

respectively. The area factors are products because the two coordinate spaces are orthogonal. Lemma 2 now gives

$$h_{\Pi(A\times B)}(u, v) = |B| h_{\Pi A}(u) + |A| h_{\Pi B}(v).$$

This is the support function of the right side of (5), proving that identity. Hence

$$\begin{aligned} R_{r+s}(A \times B) &= \frac{|B|^r |A|^s |\Pi A||\Pi B|}{(|A||B|)^{r+s-1}} \\ &= \frac{|\Pi A|}{|A|^{r-1}} \frac{|\Pi B|}{|B|^{s-1}}, \end{aligned}$$

as required. □

The product identity reduces our comparison to the values of its factors. To use simplices as a dimension-dependent benchmark, we also need to know that all full-dimensional simplices of a fixed dimension have the same value. The following affine covariance formula supplies that fact.

Lemma 4. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a convex body. For every invertible linear map $L\colon \mathbb{R}^d \to \mathbb{R}^d$,*

$$\Pi(LP) = |\det L| L^{-t} \Pi P, \qquad R_d(LP) = R_d(P),$$

where L^{-t} denotes the inverse transpose. Translations also preserve R_d.

Proof. First suppose that P is a polytope. The transformed outward unit normal of F is $L^{-t}\nu_F/\|L^{-t}\nu_F\|$. Its area is

$$s_{LF} = |\det L|\,\|L^{-t}\nu_F\|\,s_F.$$

For the area formula, take a prism of height one over F, extending in direction ν_F. Its image has volume $|\det L|s_F$ and height $1/\|L^{-t}\nu_F\|$ over LF; dividing volume by height gives the formula. Thus each area-normal vector transforms by $|\det L|L^{-t}$, including when $\det L < 0$. Apply (4). The absolute determinant of this map is $|\det L|^{d-1}$, exactly the scaling of $|LP|^{d-1}$. Translation invariance follows directly from projection volumes.

For completeness, we justify the passage to an arbitrary convex body and the existence of its projection body. Translate an interior point of P to the origin, choose $R > 0$ with $P \subset RB_2^d$, and take increasing inscribed polytopes P_m containing a fixed ball aB_2^d, $a > 0$, such that

$$P_m \subset P \subset P_m + \varepsilon_m B_2^d \subset (1 + \varepsilon_m/a)P_m, \qquad \varepsilon_m \longrightarrow 0.$$

Take, for example, convex hulls of cumulative finite nets in P and a fixed inscribed polytope containing aB_2^d. Put $\lambda_m = 1 + \varepsilon_m/a$, and for unit u write $F(u) = \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} P)$ and $f_m(u) = h_{\Pi P_m}(u)$. Projection and volume monotonicity give

$$0 \le F(u) - f_m(u) \le (\lambda_m^{d-1} - 1)\kappa_{d-1}R^{d-1}.$$

Thus $f_m \to F$ uniformly. The increasing bodies ΠP_m lie in a fixed ball and contain $\kappa_{d-1}a^{d-1}B_2^d$, so their union closure is a convex body with support function F, namely ΠP. The support inequalities also give

$$\Pi P_m \subset \Pi P \subset \lambda_m^{d-1}\Pi P_m, \qquad |\Pi P_m| \le |\Pi P| \le \lambda_m^{d(d-1)}|\Pi P_m|.$$

Likewise $|P_m| \le |P| \le \lambda_m^d |P_m|$. Applying L to the original sandwich gives the same convergence for LP_m. Passing to the limit in the polytope identity proves covariance and, by these volume inequalities, the asserted invariance of R_d. □

3 The simplex value

We now compute the value to which the product will be compared. Only one extra segment beyond a coordinate cube is needed to describe the projection body of a simplex.

Proposition 5. *For $d \ge 2$, the standard simplex $T_d = \mathrm{conv}(0, e_1, \dots, e_d) \subset \mathbb{R}^d$ satisfies*

$$|T_d| = \frac{1}{d!}, \qquad |\Pi T_d| = \frac{d+1}{((d-1)!)^d}, \qquad R_d(T_d) = c_d.$$

Every full-dimensional d-simplex has the same value c_d.

Proof. The simplex volume follows by iterated base and height. Put $w = e_1 + \cdots + e_d$. The coordinate facets have area-normal vectors $-e_i/(d-1)!$. The remaining facet has unit normal $w/\sqrt{d}$. Dropping its last coordinate projects it onto the standard $(d-1)$-simplex with Jacobian $1/\sqrt{d}$; its area-normal vector is therefore $w/(d-1)!$. In the centered segments of Lemma 2, changing the sign of a generator changes no segment. Translating each segment to start at the origin therefore shows that ΠT_d is a translate of

$$\frac{[0,1]^d + [0,w]}{(d-1)!}.$$

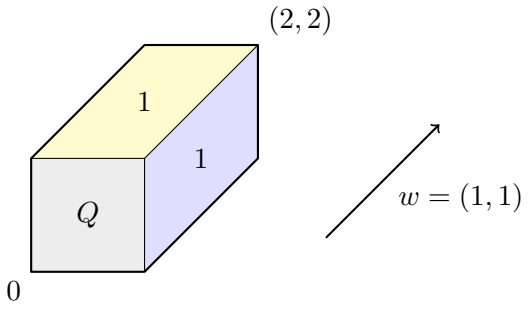


Figure 1: The planar extrusion $Q + [0, (1, 1)]$, with $Q = [0, 1]^2$, consists of the unit square and two parallelograms of area one. Its area is 3. The fiber argument in Proposition 5 proves the corresponding volume $d + 1$ in every dimension.

To compute the numerator's volume, let $Q = [0, 1]^d$. Every nonempty fiber of Q parallel to w is an interval, and adding $[0, w]$ increases its length by $\|w\|$. The base projection is unchanged. Fubini therefore gives

$$|Q + [0, w]| = |Q| + \|w\| \operatorname{vol}_{d-1}(\operatorname{proj}_{w^\perp} Q) = 1 + h_{\Pi Q}(w).$$

The cube has two facets of area one in each coordinate direction, so (3) gives $h_{\Pi Q}(w) = d$. Thus $|Q + [0, w]| = d + 1$; Figure 1 illustrates the two-dimensional case. Dilation by $1/(d-1)!$ and division by $|T_d|^{d-1}$ now yield

$$R_d(T_d) = \frac{(d+1)(d!)^{d-1}}{((d-1)!)^d} = \frac{(d+1)d^d}{d!}.$$

Every full-dimensional simplex is an invertible affine image of T_d, so Lemma 4 proves the final assertion. □

4 The counterexample and exponential excess

The geometric work is complete. The product formula turns the comparison in dimension twenty into a calculation with two simplex constants.

Proof of Theorem 1. The product $K = T_{10} \times T_{10}$ is compact and convex with nonempty interior in $\mathbb{R}^{20}$. Propositions 3 and 5 give

$$R_{20}(K) = R_{10}(T_{10})^2 = \left(\frac{11 \cdot 10^{10}}{10!}\right)^2.$$

Consequently,

$$\frac{R_{20}(K)}{c_{20}} = \frac{121 \cdot 10^{20} \cdot 20!}{(10!)^2 \cdot 21 \cdot 20^{20}} = \frac{121\binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096} > 1.$$

Here $\binom{20}{10} = 184\,756$, and the exact integer certificate is

$$121 \cdot 184\,756 - 21 \cdot 1\,048\,576 = 335\,380 > 0.$$

This is an exact strict violation of (2). □

More generally, for $r, s \geq 2$, the same two propositions give the explicit product test

$$\frac{R_{r+s}(T_r \times T_s)}{c_{r+s}} = \frac{(r+1)(s+1)}{r+s+1}\binom{r+s}{r}\frac{r^r s^s}{(r+s)^{r+s}}.$$

Thus any universal upper constant in dimension $r+s$ must be at least $c_r c_s$.

Iterating the product test gives the following quantitative consequence.

Corollary 6. *There exist a constant $\lambda > 1$ and an integer $N \geq 20$ such that, for every integer $n \geq N$, some full-dimensional convex polytope $K_n \subset \mathbb{R}^n$ satisfies*

$$\frac{R_n(K_n)}{R_n(T_n)} \geq \lambda^n.$$

The polytopes K_n may be chosen as Cartesian products of simplices.

Proof. We use repeated ten-dimensional factors to obtain exponential growth, and one factor of dimension between ten and nineteen to reach every sufficiently large dimension. For $r \in \{0, \dots, 9\}$, put $d_r = 10 + r$. Every integer $n \geq 20$ has a unique representation $n = 10k + d_r$ with $k \geq 1$. In the orthogonal coordinate product space, take

$$K_n = T_{10}^k \times T_{d_r}.$$

Here T_{10}^k denotes the k-fold Cartesian product. All factors and their successive products are full-dimensional convex polytopes, with each factor dimension at least ten. Thus Propositions 3 and 5 apply iteratively and give

$$R_n(K_n) = c_{10}^k c_{d_r}.$$

To compare this value with c_n, we bound the simplex constants by $n+1$ times a fixed exponential, whose rate is smaller than that supplied by the ten-dimensional factors. Put $\beta = 11/4$. The exponential series gives

$$e = \frac{5}{2} + \sum_{j=3}^{\infty} \frac{1}{j!} < \frac{5}{2} + \frac{1}{6}\sum_{j=0}^{\infty} 3^{-j} = \beta,$$

since the successive ratios in the tail are strictly less than $1/3$. The positive term $n^n/n!$ in the series for e^n is strictly smaller than the full sum, so

$$c_n < (n+1)e^n < (n+1)\beta^n \qquad (n \geq 2).$$

For the fixed ten-dimensional block, exact arithmetic yields

$$c_{10} = \frac{17\,187\,500}{567}, \qquad A := \frac{c_{10}}{\beta^{10}} = \frac{1\,638\,400\,000\,000}{1\,336\,956\,340\,797} > 1.$$

Choose a fixed λ with $1 < \lambda < A^{1/10}$. Combining these estimates with the formula for $R_n(K_n)$ gives

$$\frac{R_n(K_n)}{R_n(T_n)\lambda^n} > c_{d_r}\beta^{-d_r}\lambda^{-d_r}\frac{(A/\lambda^{10})^k}{10k + d_r + 1}.$$

For each fixed r, the right side tends to infinity as $k \to \infty$, because $A/\lambda^{10} > 1$. A threshold can therefore be chosen in each of the ten residue classes; their maximum, enlarged to be at least twenty, gives one N for all $n \geq N$. □

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